



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SUBRAMANYA ENTERPRISES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

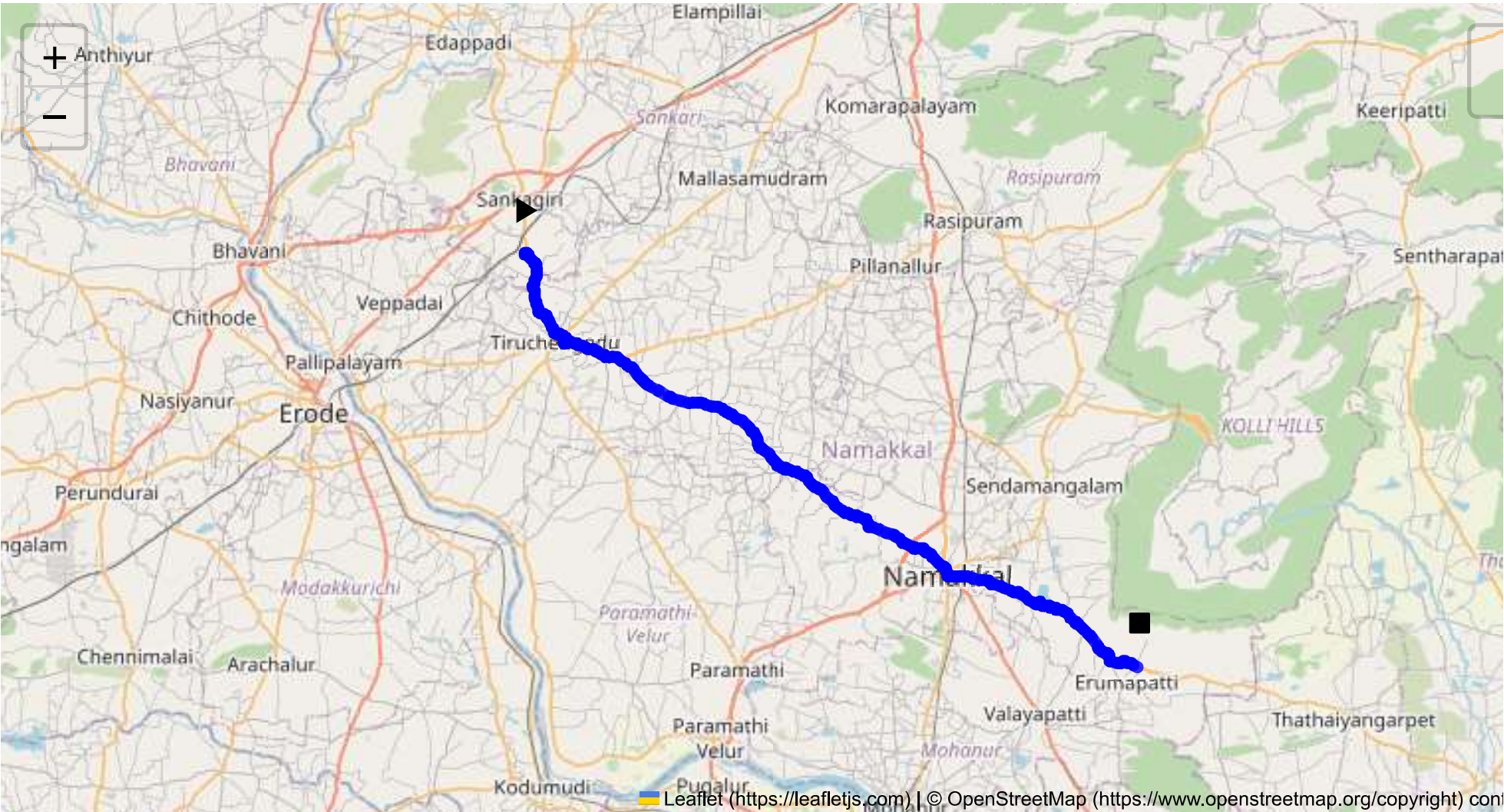
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 61.25 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (11.156951, 78.299965)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report: CVQF+23W, Sangagiri to 574X+RQJ, Erumapatti, Tamil Nadu

1. Overview of the Route Map: The route from Sangagiri to Erumapatti covers approximately 61.25 kilometers, traversing through rural and semi-urban landscapes, and is primarily accessible via National and State Highways. The journey typically follows a major highway that facilitates freight movement and connects various small towns and villages.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu generally experiences a tropical climate. The weather is predominantly hot and humid, with the peak summer months (March to May) experiencing temperatures climbing up to 40°C (104°F). The monsoon season (October to December) can bring heavy rainfall leading to waterlogging and reduced visibility. Drivers should be wary of slippery roads and potential flooding, especially near low-lying areas during these times.

3. Analysis of Traffic Patterns: Traffic can vary significantly along this route. Peak congestion is generally observed during the early morning hours (7:00 AM - 9:00 AM) and late afternoons (5:00 PM - 7:00 PM) due to school and office commutes. The highways are usually less congested during midday. Particular congestion spots include urban intersections in small towns that lack efficient traffic management.

4. Assessment of Road Quality and Infrastructure: The primary highways on this route are generally well-maintained, but truck drivers may encounter narrow lanes, potholes, and uneven road surfaces, particularly on smaller connecting roads. Some infrastructural concerns include insufficiently marked road signs and inadequate lighting in more remote stretches.

5. Suggestions for Alternative Routes for Emergencies: In emergencies, an alternative route via smaller local roads may connect back to major highways, but these should be chosen based on updated local advisories as these roads may not be suitable for heavy vehicles. It is advisable to have a navigation app with live updates to find the best available alternative path.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Hauling hazardous materials through Tamil Nadu mandates adherence to both state and national regulations, which include restrictions on transporting certain types of hazardous goods through densely populated areas and during specific hours. These rules are subject to periodic updates.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: Records from the past decade indicate a few incidents involving heavy vehicles, typically due to mechanical failures or road accidents. Historical data highlights that driver fatigue and speeding were primary causes. Fewer incidents involved hazardous materials, thanks to stringent regulatory compliance.

8. Environmental Considerations and Sensitive Areas: The route passes through agricultural zones, which are sensitive to pollution. Drivers are urged to avoid unnecessary idling of engines to reduce emissions. Spills or leaks from hazardous materials could severely impact the local ecosystem, making precautionary measures essential.

9. Analysis of Communication Coverage: Most of the route is well-covered by telecom networks, though there might be occasional dead zones, particularly in more rural and less populated sections. Drivers should ensure their communication devices are fully functional, and planned checks should be performed when entering potentially undeveloped areas.

10. Estimated Emergency Response Times for Different Route Segments: Emergency response times can vary, with faster responses typically occurring closer to urban centers where medical and emergency facilities are nearby (approximately 15-30 minutes). In more remote segments, response times could extend beyond 45 minutes.

11. Overall Summary of Risk Assessment: Overall, the route from Sangagiri to Erumapatti poses moderate risk for truck drivers, particularly for those conveying hazardous materials. Risks chiefly involve

weather-related hazards, road conditions in rural areas, and adherence to safety regulations. Emergency protocols should always be prepared, and drivers should remain vigilant, especially during peak hour traffic and inclement weather conditions. By following regulated paths and observing all safety practices, risks can be substantially mitigated.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	Medium	11.38342, 77.89975	30 KM/Hr
5	Turn	Medium	11.38295, 77.90001	30 KM/Hr
6	Turn	High	11.38186, 77.90258	15 KM/Hr
7	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
8	Turn	Medium	11.22407, 78.16640	30 KM/Hr
9	Turn	High	11.21816, 78.16934	15 KM/Hr
10	Turn	Medium	11.20105, 78.23198	30 KM/Hr
11	Turn	High	11.16124, 78.28095	15 KM/Hr
12	Turn	Medium	11.15975, 78.28885	30 KM/Hr
13	Turn	Medium	11.16047, 78.28943	30 KM/Hr
14	Turn	Medium	11.16056, 78.28971	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
16	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
19	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
20	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
21	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
22	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
23	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
24	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
25	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
26	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
27	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
28	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
29	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
31	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
32	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
33	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
34	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
35	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
36	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
37	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
38	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium
39	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
40	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
41	hospital	Nirmala Hospital	11.2179361, 78.1665001	30 km/h	Medium
42	hospital	Shanthi Hospitals, Namakkal	11.2190661, 78.1685735	30 km/h	Medium
44	police	Police Station	11.2185353, 78.1695347	30 km/h	Medium
45	police	Women Police Station	11.2183133, 78.1692677	30 km/h	Medium
46	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
47	hospital	Government Hospital	11.2168924, 78.169068	30 km/h	Medium
48	hospital	Ganga Hospital	11.21742, 78.170272	30 km/h	Medium
49	hospital	SMK Hospital	11.2172759, 78.1706866	30 km/h	Medium
50	hospital	Manoharan Hospital	11.218016, 78.171629	30 km/h	Medium
51	hospital	Saravana Hospital	11.216652, 78.171298	30 km/h	Medium
52	hospital	Latha Eye Hospital	11.2185689, 78.1719035	30 km/h	Medium
53	hospital	Vasan Eye Care Hospital, Namakkal	11.216301, 78.1717694	30 km/h	Medium
54	hospital	Aishwarya Hospital	11.2164042, 78.1715972	30 km/h	Medium
55	hospital	Poornima Hospital	11.215533, 78.16911	30 km/h	Medium
56	hospital	Nagarajan Hospital	11.2171673, 78.1718618	30 km/h	Medium
57	clinic	Ramesh ENT & Skin Care	11.217133, 78.171071	30 km/h	Medium
58	clinic	ENT Clinic	11.215786, 78.170545	30 km/h	Medium
59	clinic	Hari Chest Clinic	11.2162053, 78.1710093	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
60	hospital	Dr. Kandasamy Hospital	11.216051, 78.172052	30 km/h	Medium
61	clinic	Dr. Shanmugam ENT Clinic	11.218762, 78.171206	30 km/h	Medium
62	clinic	Children's Clinic	11.217004, 78.171504	30 km/h	Medium
63	clinic	Krishiv's Dental Care & Orthodontic Centre	11.2184886, 78.1713346	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
17	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
18	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
30	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium